

Blinded Inverse Takeoff

Finite-horizon limits in recovering recursive redundancy

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Abstract. Memoization collapses repeated calls in a recursion tree into shared states. The velocity takeoff profile records how quickly that redundant work appears as input size grows. This note studies the inverse question: can a finite, noisy takeoff trace reveal the recurrence that produced it? We give an exact finite-horizon obstruction. For every observation length there are distinct aperiodic nonnegative-integer recurrences with the same terminal tree speed, the same total branching, the same polynomial memoized-state growth, and an identical observed takeoff prefix. We then report two blinded AWS sweeps containing 1.16 million hidden schemas and 32.67 million recovery trials. A generic matrix-pencil baseline succeeds on many visible short-lag recurrences but fails sharply on the constructed hidden-lag family. Under noise, accurate forecasting often survives after modal identification has collapsed. The result is both a negative theorem and a practical warning: predicting a recursive response is not the same as identifying its mechanism.

1 Why invert a takeoff curve?

A naive recursive program exposes a tree of calls. Equal labels can appear many times; memoization identifies them and leaves a directed acyclic graph (DAG) of distinct subproblems. Let N_n be the number of calls in the naive tree on input size n , and let M_n be the number of distinct reachable labels. The logarithmic overlap is

$$S_n = \log \frac{N_n}{M_n},$$

with one-step velocity

$$v_n = S_{n+1} - S_n = u_n - w_n,$$

where

$$u_n = \log N_{n+1} - \log N_n, \quad w_n = \log M_{n+1} - \log M_n.$$

If N_n grows exponentially with speed $\alpha > 0$ while M_n is polynomial, then $u_n \rightarrow \alpha$, $w_n \rightarrow 0$, and $v_n \rightarrow \alpha$. That terminal number says how fast redundancy eventually accumulates, but not how it gets there. Two recurrences can have the same cruising speed while one accelerates immediately and the other rings for a very long time.

The normalized takeoff profile and its causal difference are

$$F_n = \frac{u_n}{\alpha}, \quad \kappa_n = F_n - F_{n-1}, \quad F_{-1} = 0.$$

Thus $F_n \rightarrow 1$, and κ records the response shape: delay, monotone settling, alternating correction, or damped ringing. Forward analysis starts from a recurrence and predicts F . Inverse takeoff starts from an observed prefix of F and asks what can be recovered.

The analogy is system identification, but the object is unusual. The “plant” is the redundant response of a recursion tree before memoization quotients equal states. Classical Prony and matrix-pencil methods can estimate damped exponential modes from samples [1, 2, 3]. What is new is the constrained source class and the distinction between response modes and literal recursive branches.

2 Finite-lag response and its modes

Consider the finite-lag schema

$$N_n = 1 + \sum_{j \in J} a_j N_{n-j}, \quad N_m = 1 \text{ for } m \leq 0, \quad (1)$$

where $J \subset \{1, 2, \dots\}$ is finite and $a_j \in \mathbb{Z}_{>0}$. Define

$$Q(z) = 1 - \sum_{j \in J} a_j z^j.$$

When $\gcd(J) = 1$ and $\sum_j a_j > 1$, Q has a unique smallest-modulus positive root $\rho \in (0, 1)$, and the terminal speed is $\alpha = -\log \rho$. Non-dominant singularities ζ contribute normalized correction modes

$$\theta = \frac{\rho}{\zeta}.$$

The affine $+1$ in (1) also contributes the singularity $z = 1$. The largest active $|\theta| < 1$ controls the eventual settling rate, while a complex leading mode produces ringing.

For a common terminal speed $\alpha = \log 2$, the root is $\rho = 1/2$ and the coefficient constraint becomes

$$\sum_j a_j 2^{-j} = 1. \quad (2)$$

This dyadic identity is useful experimentally: it generates many structurally different recurrences while holding the first-order asymptotic invariant exactly fixed.

An observed tail is modeled approximately as

$$Y_n := F_n - 1 \approx \sum_{r=1}^m c_r \theta_r^n. \quad (3)$$

The approximation matters. Even when N_n has a finite exponential expansion, the logarithmic ratio defining F_n introduces higher-order mode products. A low-rank spectral fit is therefore a tail model, not an algebraic identity for every finite prefix.

Figure 1 gives a visual vocabulary for the inverse problem. The top row is the takeoff profile; the bottom row is its causal

difference, the kernel actually carrying the transient shape. The same terminal level can be approached smoothly, by sign alternation, through damped oscillation, or in widely separated echoes.

3 An exact finite-horizon obstruction

The central difficulty is more basic than numerical noise: the recursive signal may not yet have returned from its hidden lag.

Proposition 1 (Fixed-invariant finite-horizon nonidentifiability). *Fix an integer observation length T . There exist distinct aperiodic finite-lag schemas satisfying (2), having the same total branch count and $M_n = O(n)$, whose takeoff profiles agree exactly on $F_0, \dots, F_{T-1}$.*

Proof. Choose $L \geq T$ and an integer $1 \leq t < 2^{L+1}/3$. Set

$$a_L = t, \quad a_{L+1} = 2^{L+1} - 3t, \quad a_{L+2} = 2t.$$

All three coefficients are positive. Their total is $A = 2^{L+1}$, independent of t , and

$$4a_L + 2a_{L+1} + a_{L+2} = 2^{L+2},$$

which is (2) after division by 2^{L+2} . Hence every member has $\alpha = \log 2$. The consecutive support $\{L, L+1, L+2\}$ is aperiodic, and its reachable labels lie in a one-dimensional interval, so $M_n = O(n)$.

For $1 \leq n \leq L$, every referenced boundary value in (1) equals one. Therefore

$$N_1 = \dots = N_L = 1 + A$$

for every t . It follows that $F_0, \dots, F_{L-1}$ are identical. At the next step,

$$N_{L+1} = 1 + A + tA,$$

so different t values separate. They also define distinct polynomials Q_t ; because each has constant term one, their full root multisets cannot all coincide. Thus no estimator receiving only the common prefix can recover the coefficient vector correctly for every member. $\square$

Intuitively, this is like listening for an echo before sound has reached the reflecting wall. Unlimited estimator sophistication cannot recover information that is absent from the observation window. The proposition concerns distinct combined recurrences, not merely different stories that split the same coefficients into latent channels. Figure 2 shows the construction directly: different hidden branch allocations have the same fixed-speed invariant and exactly the same visible prefix, then emit different echoes when the longest lag begins to return.

Table 1: Completed AWS runs.

Run	Schemas	Trials	Time
Broad baseline	575,440	13,810,560	18.6 min
Horizon sweep	589,302	18,857,664	14.2 min
Total	1,164,742	32,668,224	32.8 min

4 The blinded experiment

The study used five fixed-speed families: immediate binary recursion; the near-periodic slow family $a_L = 2^L - 1$, $a_{L+1} = 2$; the proposition’s finite-horizon family; a two-scale family with one local and one delayed channel; and random dyadic refinements that replace one branch at lag j by two branches at lag $j + 1$. Every such refinement preserves (2).

For each hidden schema we computed an exact integer call-count sequence and formed F_n . The estimator received only

$$\tilde{Y}_n = Y_n + \sigma \max\{\text{RMS}(Y), 10^{-12}\} \xi_n, \quad \xi_n \sim \mathcal{N}(0, 1),$$

not the coefficients or true roots. We tested windows beginning at the origin and immediately after the maximum lag. Relative noise levels were 0, 10^{-8} , 10^{-5} , 10^{-3} .

The baseline formed Hankel matrices

$$(H_0)_{ij} = \tilde{Y}_{i+j}, \quad (H_1)_{ij} = \tilde{Y}_{i+j+1},$$

selected an order from one through six by held-out prediction, and recovered poles from a truncated-SVD matrix pencil. It then forecast 24 unseen samples. We scored the leading-pole radius, qualitative class (monotone, alternating, or ringing), and holdout normalized RMSE. “Forecast success” means NRMSE below 0.1. Ground truth was used only after fitting.

Two independent seeded sweeps were run. Table 1 reports launcher wall time, which slightly overstates billable instance time. At the verified rate of \$1.428/hour, estimated compute was \$0.782 total. Both instances terminated and all temporary cloud resources were removed.

The broad run used lengths 24, 48, 96 and lags through 18. The horizon sweep used lengths 8, 16, 32, 64 and lags through 32, permitting explicit bins in T/L . Each schema contributed every window, regime, and noise condition; the schema generator itself was stochastic rather than an exhaustive uniform census.

5 Results

Figure 3 summarizes the two most important empirical boundaries: information arriving after the hidden lag, and prediction remaining reliable after modal identification degrades.

Sub-lag observations are blind.. Table 2 isolates noiseless finite-horizon traces beginning at the origin. Modal-class accuracy was zero throughout $T/L < 1$. This empirical zero is consistent with the proposition, although the theorem itself proves coefficient nonidentifiability rather than a particular classifier’s error. There is no rapid recovery immediately after

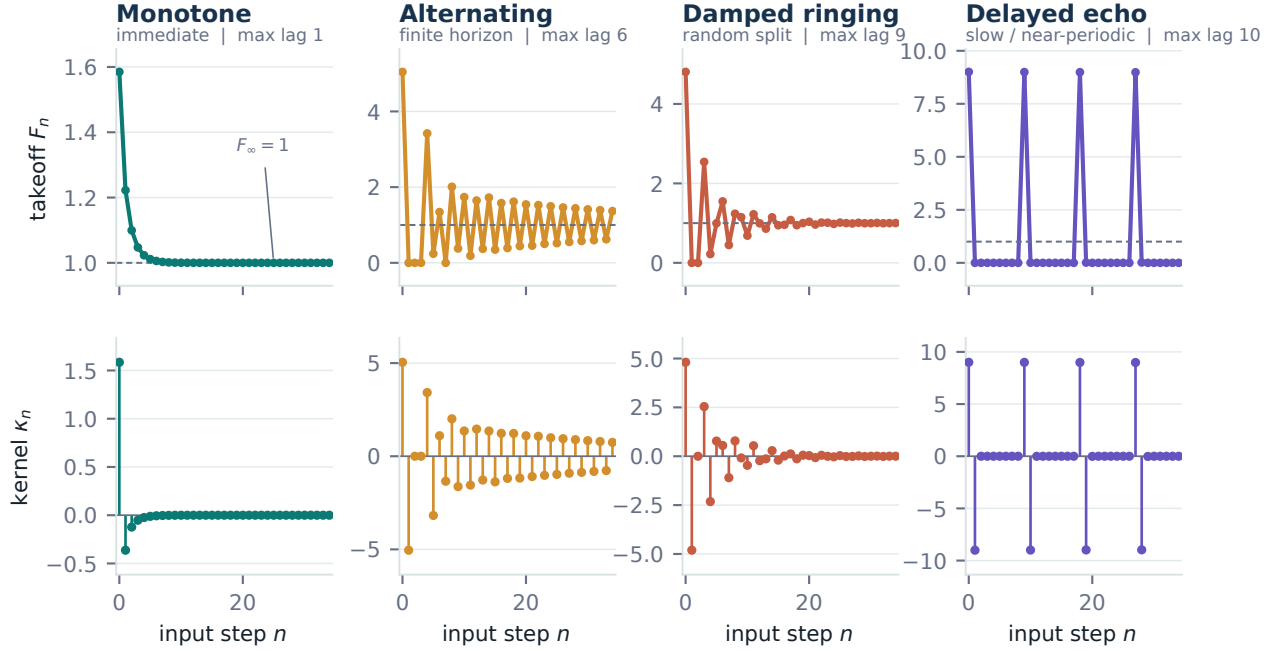


Figure 1: A kernel atlas from representative experimental schemas. All four recurrences have terminal speed $\alpha = \log 2$ and hence $F_n \rightarrow 1$, but their finite responses are plainly different. The dashed line marks the common terminal level. The lower row shows $\kappa_n = F_n - F_{n-1}$: a short relaxation, alternating correction, damped ringing, and a near-periodic echo train. Different vertical scales are used so that each transient remains legible.

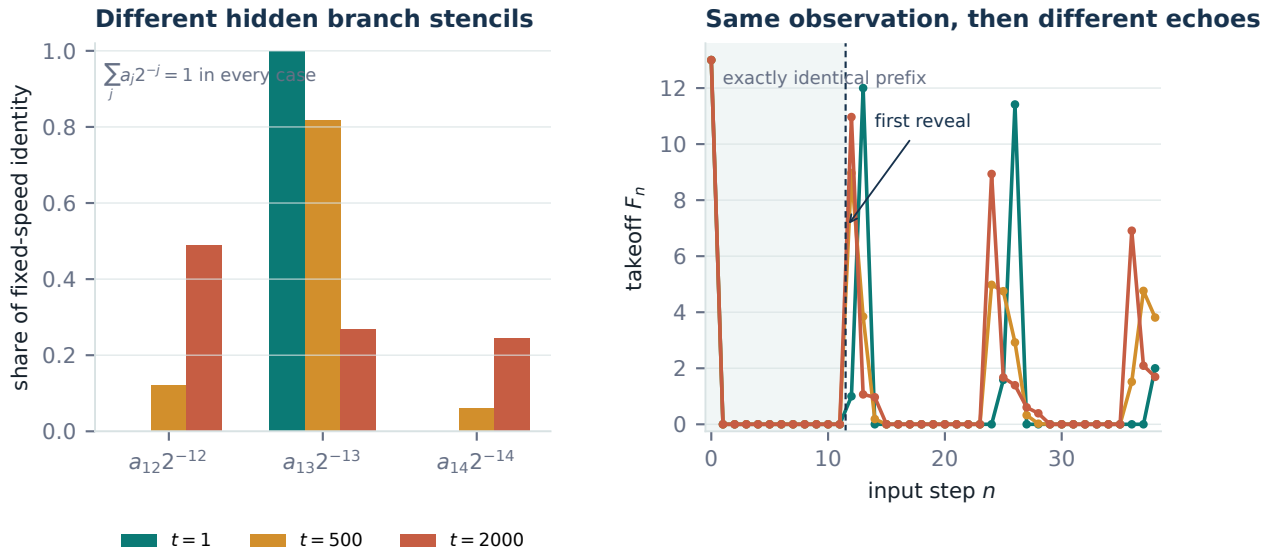


Figure 2: Exact finite-horizon blindness at $L = 12$. Left: three distinct coefficient stencils, plotted by their contribution $a_j 2^{-j}$ to the fixed-speed identity. Right: their takeoff profiles coincide exactly on $F_0, \dots, F_{11}$ (shaded) and first separate at F_{12} . Later pulse trains retain visibly different recursive signatures. This collision is algebraic, not a consequence of noise or an unsuccessful optimizer.

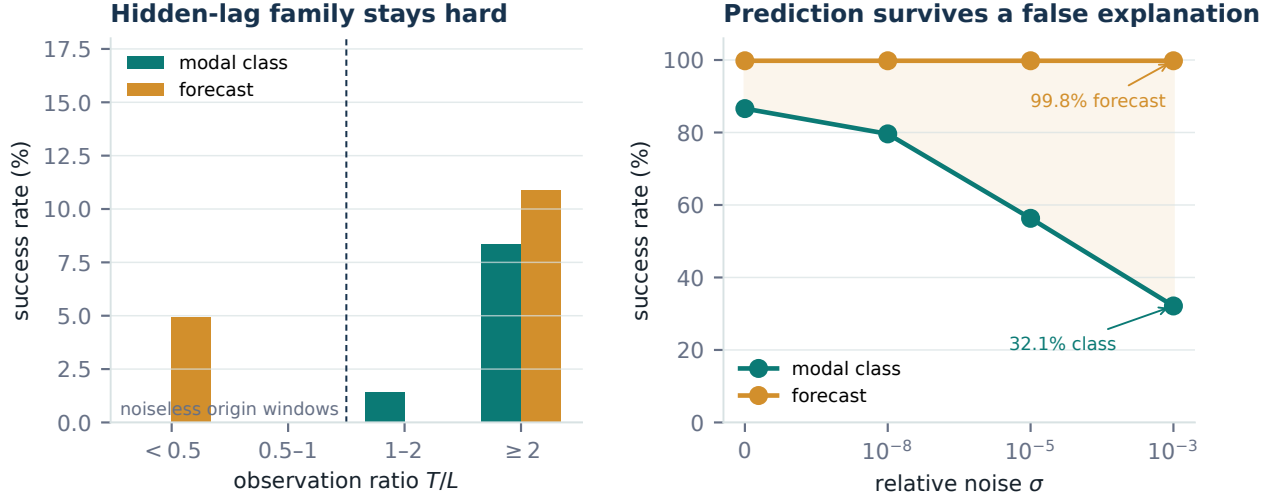


Figure 3: Recovery phase diagrams from the completed AWS sweeps. Left: on the constructed finite-horizon family, noiseless origin-window recovery is absent below the hidden lag and remains weak after it becomes visible. Right: on 144,863 post-lag random-split schemas with a 96-sample window, modal classification degrades sharply with relative noise while 24-step forecast success remains 99.8%. Bars and curves are computed directly from the saved run summaries.

Table 2: Noiseless finite-horizon family, origin windows.

T/L	Trials	Class acc.	Forecast succ.
< 0.5	101,309	0.0%	4.9%
[0.5, 1)	132,303	0.0%	0.0%
[1, 2)	147,761	1.4%	0.0%
≥ 2	209,671	8.4%	10.9%

the lag: the generic six-mode baseline remained poor even for $T/L \geq 2$. These schemas can have degree 34 and crowded near-periodic modes, so the post-lag failure is an empirical limitation of the baseline, not a second impossibility theorem.

The estimator is not universally ineffective.. Random dyadic-split recurrences were much easier once observations began after the largest lag. In the noiseless broad run, class accuracy was 78.3%, 88.9%, and 86.6% for windows 24, 48, and 96; forecast success was 95.6%, 99.5%, and 99.8%. The corresponding sampled median leading-radius errors were 0.0024, 0.0007, and 0.0005. Thus the hard-family failure is structurally selective.

Prediction can survive a false explanation.. For post-lag random-split traces of length 96, raising relative noise from zero to 10^{-3} reduced modal-class accuracy from 86.6% to 32.1% and increased sampled median radius error from 0.0005 to 0.4453. Forecast success nevertheless remained 99.8%. A finite-window predictor can therefore extrapolate accurately while assigning the wrong modal geometry. This is the most practically important empirical warning: predictive fit, modal identification, and source identification are three different evidentiary levels.

6 What the experiment establishes

The exact contribution is modest but sharp. Without a lag bound, finite-prefix source recovery is impossible even after terminal speed, total branching, DAG growth class, and aperiodicity are fixed. The experiment then shows where a standard spectral tool works and where it does not. Massive replication makes the family contrast stable; it does not turn the classical matrix-pencil algorithm into a novel method.

Several limitations matter. The schemas are synthetic and one-dimensional; real program traces may contain input-dependent branching and measurement artifacts. The modal baseline is capped at order six, while the hardest recurrences have much larger degree. Root calculations for high-degree, near-periodic polynomials are themselves conditioned problems. The logarithmic takeoff profile is only asymptotically a finite exponential sum, and very small late corrections approach floating-point resolution. Finally, millions of trials do not remove bias from the chosen schema distribution.

The next positive result should exploit information the baseline ignores:

$$a_j \in \mathbb{Z}_{\geq 0}, \quad \sum_j a_j 2^{-j} = 1,$$

together with a declared lag bound and sparsity. A constraint-aware decoder can enumerate or optimize over feasible dyadic recurrences, forward-simulate their takeoff curves, and compare them on held-out samples. The academically useful test is whether those constraints shift the recovery boundary as a function of T/L , noise, pole separation, and modal radius.

Real recursive traces should then test whether the recovered equivalence classes remain stable outside the synthetic model.

7 Conclusion

Recursive redundancy casts a modal shadow, but a short shadow does not uniquely reveal the object that cast it. Before the longest lag returns, distinct fixed-speed recurrences can be exactly observationally identical. Afterward, generic modal recovery may work well on ordinary short-lag families yet remain fragile on nearly periodic ones. Most strikingly, excellent forecasts can coexist with incorrect modes. Inverse takeoff should therefore report what the data support in order: response prediction first, modal structure second, and literal recursive mechanism only under explicit identifying constraints.

References

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